Decoding Esophageal Cancer Subtypes for Precision Treatment

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1 Introduction

Cancer is a group of diseases characterized by the uncontrolled growth and spread of abnormal cells in the body. Not only is it the second-leading cause of deaths world-wide, but it also significantly reduces the quality of life of cancer patients. Esophageal cancer (ESCA) is the cancer that occurs in the esophagus, anywhere between the back of the throat to the stomach. It is the seventh most common cause of cancer deaths. There are two main subtypes of esophageal cancer: esophageal carcinoma (EAC) and esophageal squamous-cell carcinoma (ESCC). Although both subtypes occur in the esophagus, their underlying molecular mechanisms are significantly different. Therefore, the subtypes require different treatment strategies. Therefore, identifying the cancer subtype as early as possible is crucial for designing a suitable medical intervention plan, leading to better clinical outcomes.

The objective of this project was to build machine learning classifiers which are trained to classify esophageal cancer patients into the two subtypes: EAC and ESCC. There have been several studies which have identified miRNAs as important biomarkers in the early diagnosis of esophageal cancer[1−4]. There are some studies which have identified several miRNAs that are differentially expressed in EAC and ESCC[5−7]. Therefore, miRNA-seq counts were used as features for training our classifiers. After training the classifiers, the model performances were evaluated and compared to find the best model for this dataset.

In this project, three machine learning classifiers were trained: Support Vector Machine(SVM) Classifier, Random Forest Classifier and Logistic Regression. These three models were chosen models for this project because these models have been successfully used in related works[8,9]. Furthermore, these three models are the most used models in the context of The Cancer Genome Atlas (TCGA) data[10]. SVM Classifier is known to be effective in high dimensional spaces, especially when the number of features exceeds the number of samples. Random Forest Classifier is known to be efficient with large datasets. It doesn't overfit for datasets with large numbers of features. Logistic Regression is a simple and easily interpretable model. It can also be extended to multiple classes, which could be useful to add more cancer subtypes in the analysis. Both Random Forest Classifier and Logistic Regression provide an estimate of the significance of the features.

The TCGA-ESCA data was used for this project, which is publicly available on National Cancer Institute's Genomic Data Commons (GDC) portal. This dataset contains genomic and clinical data of patients with esophageal cancer. The TCGA-ESCA data includes genomic data such as sequencing

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reads, transcriptome profiling data, DNA methylation data, as well as clinical data such as patient age, sex, cancer stage, and treatment information.

2 Data

For this project, the classifiers were trained on the miRNA expression data of the patients. Since the molecular mechanisms driving the two cancer subtypes are different, different miRNAs might be dysregulated in the patients with different subtypes. Using miRNA expression data as features will enabled us to exploit these differences and train the models. Since the features represent the expression levels of different miRNAs, they are continuous variables. The normalized miRNA expression data available in the TCGA-ESCA data on the GDC portal were used for the project.

The dataset created from TCGA-ESCA can be described as follows:

- Number of samples - 184
  - EAC patient samples - 88
  - ESCC patient samples - 95
- Number of features - 1881

The data was split as 80:20 to generate the train-test datasets.

3 Methods

Described below are the various methods that have been used in our project.

3.1 PCA Visualization

Since the dataset was high-dimensional, PCA used to visualize the data onto two-dimensional space in order to visualize it. Here, PCA was implemented from scratch. The first step before performing PCA was to mean-center the data, the data was first standardized as follows:

$$Z = \frac{X - \mu}{\sigma}$$

After standardizing the data, the covariance matrix of the features was computed as follows:

$$A = Z^T Z$$

Then the eigenvalues and eigenvalues of the covariance matrix A were computed by using the equation:

$$A\lambda = \lambda v$$
$$(A - \lambda)v = 0$$

The equation above can be solved by solving $|A - \lambda| = 0$, which gives the eigenvalues. The corresponding eigenvectors can be computed by plugging the eigenvalues into the equation above. After these computations, the eigenvalues and corresponding eigenvectors were sorted in the order of decreasing eigenvalues. In order to visualize the data to a two-dimensional space, we projected the data onto the eigenvector space as follows:

$$X_{PCA} = X.v$$

Finally, the top two principal components form X_{PCA} were selected to be plotted as a scatter plot.

3.2 Support Vector Machine Classifier

The scikit-learn package was used for this method. SVM classifies the data points by finding the best hyperplane that separates the two classes. Even though SVM and logistic regression both try to find the best hyperplane, logistic regression is a probabilistic approach whereas SVM is based on statistical approaches. SVM finds the best hyperplane by finding the maximum margin between the hyperplanes, i.e. the margin which maximizes the distances between the two classes. The regularization parameter C was set to 1.0. The kernel used was the default ‘Radial Basis Function’. The degree of polynomial kernel function was set to 3. The kernel coefficient gamma was set to ‘scale’.

3.3 Logistic Regression

This classifier was implemented from scratch. A gradient descent algorithm was used to learn the regression coefficients of the logistic regression. Then, a probabilistic approach was employed to classify the samples into either label 0 or 1. Here, 0 refers to the adenocarcinoma subtype and 1 refers to the squamous cell carcinoma subtype. The gradient descent algorithm runs for 100 iterations. In each iteration, the algorithm first finds the gradient, that is, the derivative of the loss function and then updates each regression coefficient using the gradient calculated.
Loss function:

$$\ell(w) = \frac{1}{n} \sum_{i=1}^{n} \left[\log(1 + e^{\langle w, x_i \rangle}) - y_i \cdot \langle w, x_i \rangle \right]$$

Derivative of loss function with respect to w^j where j ranges from 1 to number of features+1:

$$\frac{\partial \ell(w)}{\partial w^j} = \frac{1}{n} \sum_{i=1}^{n} \left[\frac{1}{(1 + e^{-\langle w, x_i \rangle})} - y_i \right] \cdot x_i^j$$

Update statement for each regression coefficient using the gradient calculated:

$$w^j \leftarrow w^j - \eta \cdot \frac{\partial \ell(w)}{\partial w^j}$$

Here, η is the learning rate.
The vector of regression coefficients is denoted by w and the length of this vector is equal to the number of features in our dataset which is 1881 plus 1 to account for finding the β_0 coefficient. So in total, the algorithm finds the values of 1882 regression coefficients for our given dataset. The derivative of the loss function or the gradient obtained is also a vector of length 1882.

Gradient descent algorithm:

Learning rate $\eta > 0$
Initialize w
While True
Compute gradient $\frac{\partial \ell(w)}{\partial w}$
If number of iterations > 100 then break
$w \leftarrow w - \eta \cdot \frac{\partial \ell(w)}{\partial w}$
End While

Here, gradient descent algorithm can be used since the negative log likelihood is being used as the loss function. So there exists a unique minimizer as it is a convex optimization problem to minimize this loss function. Once the regression coefficients were finalized after 100 iterations, the below formulas were used to get the probabilistic classification of test samples.
Formula to get probability of each class label for each sample x_i

$$P(Y = 1|x_i) = \frac{1}{(1 + e^{-\langle w, x_i \rangle})}$$

$$P(Y = 0|x_i) = 1 - P(Y = 1|x_i)$$

Based on the maximum probability obtained for each class for each sample, the algorithm assigned the class to the sample from which it gets the maximum probability for that particular sample.

3.4 Random Forest

The scikit-learn package was used for this method. A random forest is a meta estimator that fits a number of decision tree classifiers on various sub-samples of the dataset and uses averaging to improve the predictive accuracy and control over-fitting. The parameter max_depth that controls the depth of the tree was set to 2 randomly. The random state was set to 0. All the other parameters were set to their default values. Therefore, by default, the number of trees in the forest was 100. The bootstrap parameter was set to True by default which means that bootstrap samples were used to build the trees and not the whole dataset. Bootstrapping is a technique used to reduce the variance of a statistical learning algorithm. It involves creating multiple random subsets of the training data with replacement and training a model on each subset. The resulting models are then combined to produce a final prediction. This approach is called bagging or also known as, bootstrap aggregating.

4 Results

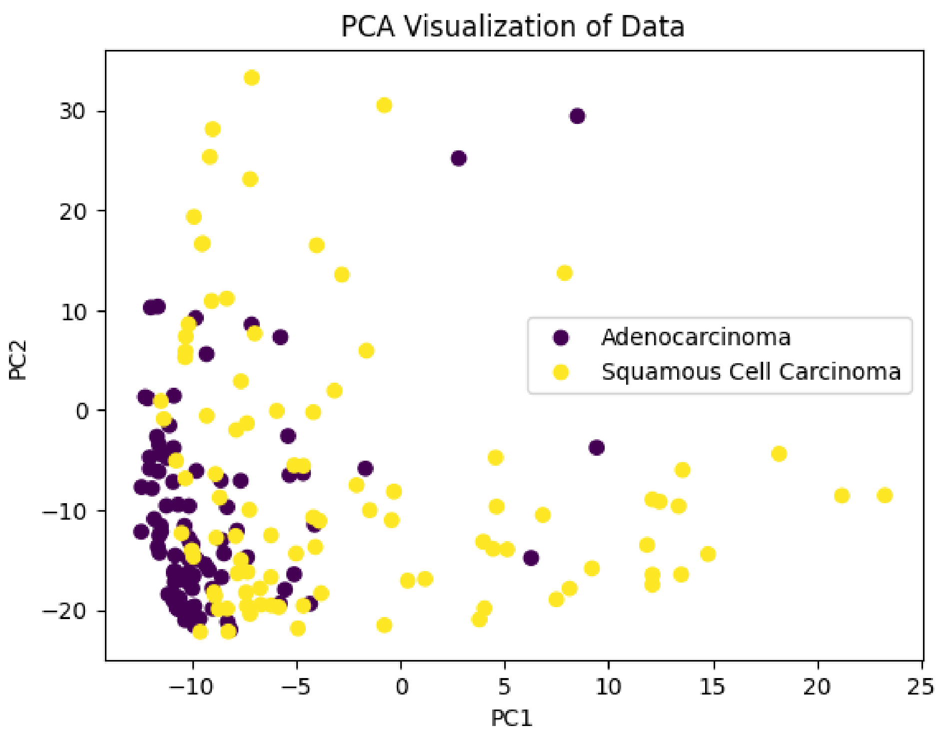

Figure 1: PCA visualization of the data in two-dimensional space.

Figure 1 shows the PCA visualization of the dataset in two-dimensional space. This plot was generated by entering the samples and their labels into the PCA_visualize function. The ESCC data points appear to be more spread out compared to the EAC data points in the 2-D space. The first two principal components explain 64.75% of the total variance.

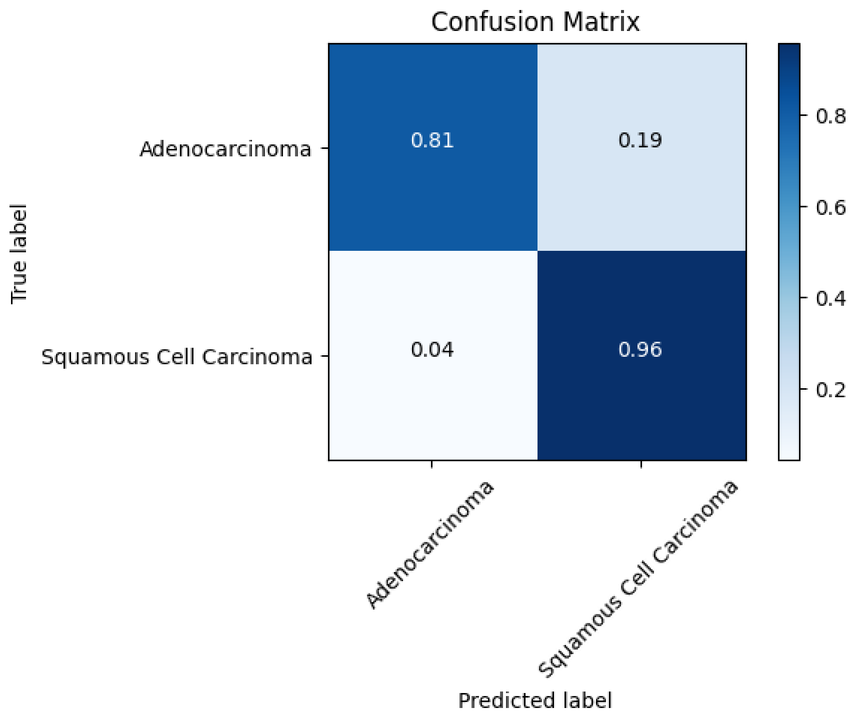

Figure 2: 10-fold Cross-validation confusion matrix for SVM Classifier

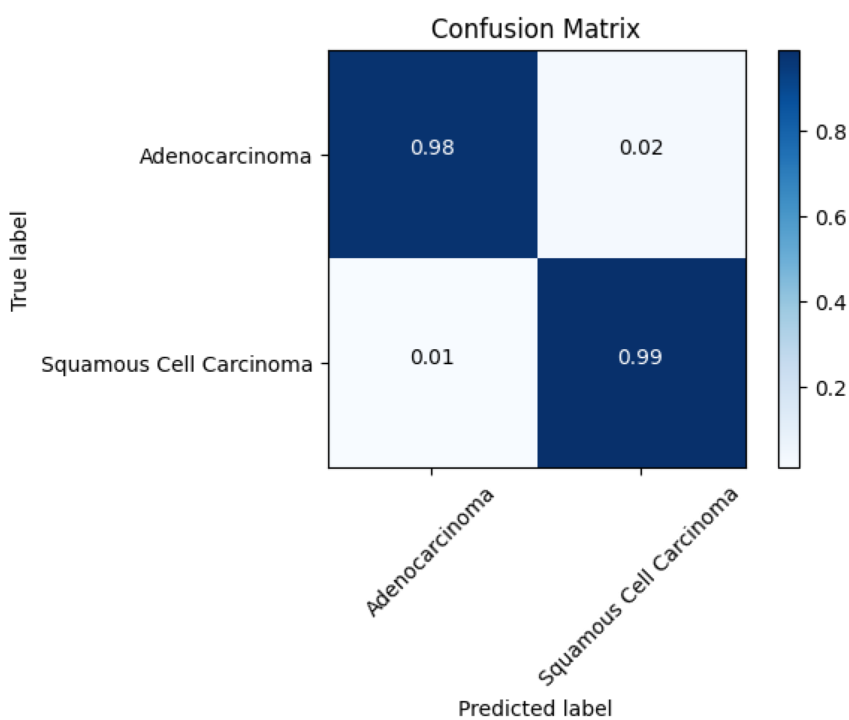

Figure 3: 10-fold Cross-validation confusion matrix for Random Forest Classifier

Fig. 2,3,4 shows confusion matrices of cross validation results obtained for each classifier. The scikit-learn function for confusion matrix calculation was used. But, additional code was written to normalize the matrices as well as change a few visual aspects of the matrices. The most correct

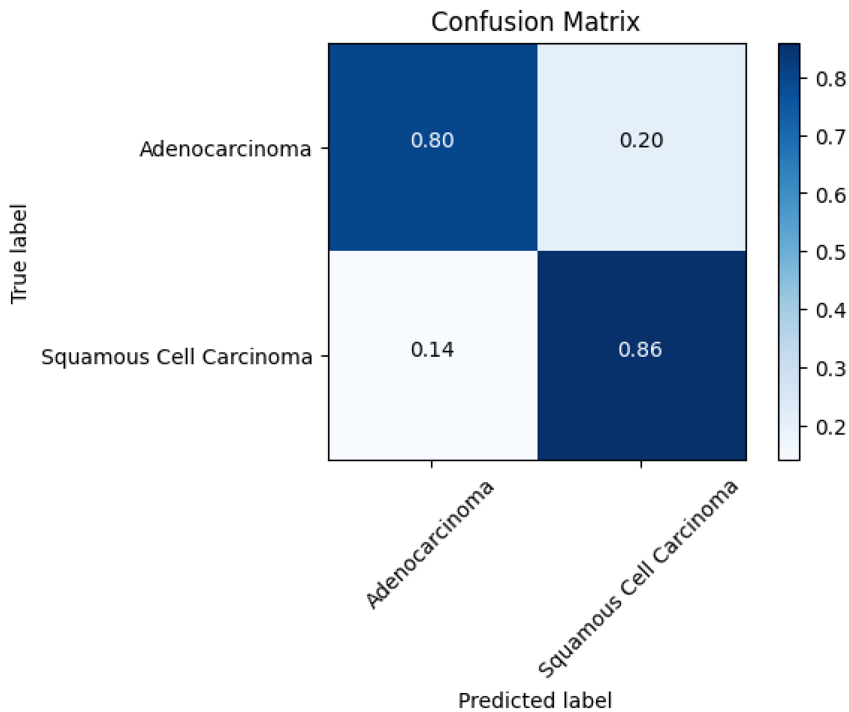

Figure 4: 10-fold Cross-validation confusion matrix for Logistic Regression

Table 1: Comparison of cross validation and test results across all three classifiers

	Cross-val accuracy	Cross-val F1-score	Test accuracy	Test F1-score
Logistic Regression	82.95%	0.83	75.68%	0.74
Support Vector Machine	88.51%	0.89	97.30%	0.97
Random Forest	97.84%	0.98	97.30%	0.97

predictions are seen in the confusion matrix of the random forest classifier. Random forest is able to detect 98% of true positives and 99% of true negatives. SVM does a good job of detecting true negatives (96%) but considerably less performance is seen in detecting true positives (81%) compared to random forests. Logistic regression has a comparable performance when compared with SVM in terms of detecting the true positives (80%). But it performs much worse when it comes to detecting true negatives (86%). Again, with the test dataset it is seen that random forest performs the best among all the three.

Table 1 shows the results of train accuracies and F1-scores and test accuracies and F1-scores. To obtain the train accuracy and F1-score of each classifier, the training dataset was trained and validated using k-fold cross validation. The k was set to 10. K-fold cross-validation is a technique used in machine learning to evaluate the performance of a predictive model. It involves partitioning the available data into K equally sized subsets or folds, and then using K-1 folds for training the model and the remaining fold for testing the model. This process is repeated K times, with each fold serving as the test set once, and the average performance across the K iterations is reported as the overall model performance. The main advantage of k-fold cross-validation is that it provides a more accurate estimate of model performance than other simpler methods, such as training on a single random split of the data. This is because it uses all the available data for training and testing, and ensures that the model is evaluated on data that it has not seen during training. To obtain the test accuracy and F1-score of each classifier, the available scikit-learn functions for accuracy and F1-score were used. Only the test dataset was given as input in this part. The results show the following trends. Random forest gives the best cross-validation results, followed by SVM and then logistic regression, both in terms of mean accuracy and F1-score. With respect to the test accuracy and F1-score, random forest and SVM perform equally good and considerably better than logistic regression.

5 Conclusion

The following conclusions have been made based on the results obtained. Random forest classifier performs the best classification for the TCGA-ESCA data. Next, best performance is shown by Support vector machine classifier.

Random forest and SVM show better performance than logistic regression as logistic regression is limited in its capacity to predict non-linear data well. Whereas, with both random forest and SVM, non-linearly separable data can be used as random forest has no such limitations and SVM uses a higher dimensional space where the data becomes linearly separable. A comparable performance is seen between random forest and SVM in terms of test accuracy and F1-score, but random forest outperforms SVM in terms of the cross-validation accuracy and F1-score. This might be because SVM classifer is a single classfier but random forest is a meta classifier as it is a number of decision trees and uses the bagging approach and so it gives the majority vote result of all the decision trees. This way the model is trained much well in the case of random forest. Another possibility of why the test accuracy is much better than the cross-validation accuracy for SVM classifier is that the test set might be more representative of the data distribution that the model was trained on.

For future work, feature selection can be done as the number of features are a lot more than the number of samples in the dataset. Next, the data distribution of the train and test set can calibrated in different ways to observe changes in cross-validation and test results especially for the SVM classifier. Lastly, other different classifiers can also be evaluated using this dataset.

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